

Task 4: Sales Prediction

Project Overview

This project predicts product sales using a Machine Learning regression model. It analyzes advertising expenditure across TV, Radio, and Newspaper media to estimate future sales.

Objective

Build a Linear Regression model to accurately predict sales based on advertising budgets.

Technologies Used

- Python
- Pandas
- Matplotlib
- Scikit-learn
- Google Colab

Workflow

- Import the required libraries.
- Load the Advertising dataset.
- Explore and preprocess the data.
- Split the dataset into training and testing sets.
- Train the Linear Regression model.
- Evaluate the model using MAE, MSE, and R^2 Score.
- Visualize actual and predicted sales.
- Predict sales for new advertising data.

Result

The Linear Regression model successfully predicts sales with good accuracy. Evaluation metrics and visualizations demonstrate the relationship between advertising expenditure and sales.